Matrix to Vine

Incomplete Pearson matrix

	X_1	X_2	X_3	X_4	X_5
X_1	1.00	−0.59	?	0.31	−0.37
X_2	−0.59	1.00	0.54	−0.79	?
X_3	?	0.54	1.00	−0.06	?
X_4	0.31	−0.79	−0.06	1.00	?
X_5	−0.37	?	?	?	1.00

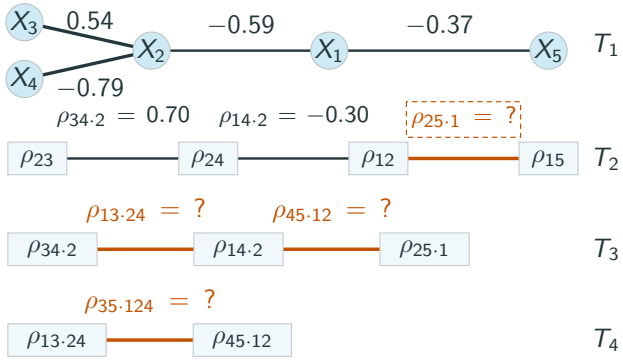
Problem

Numerical repair ensures positive semidefiniteness, but starting values and adjustment rules can introduce unjustifiable bias.

Partial correlation vine expansion

Greedy expansion – Judgement 1

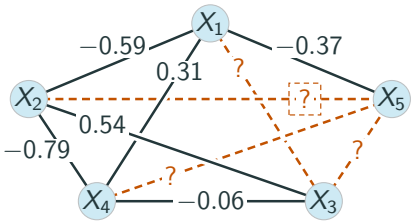
Known information first; then strongest $|\rho|$.



■ Known / derived ■ Uncertain / completion ■ Judgement

Matrix to Vine

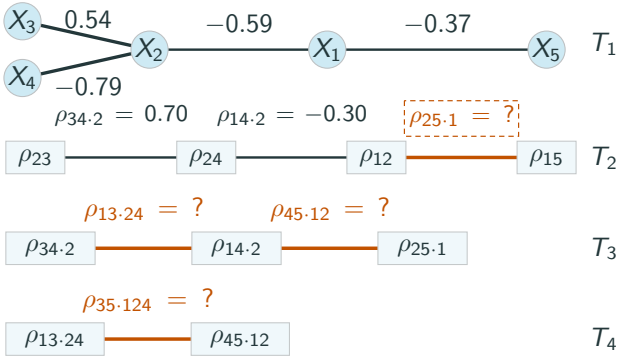
Incomplete Pearson matrix



Partial correlation vine expansion

Greedy expansion – Judgement 1

Known information first; then strongest $|\rho|$.



Vine Space Structure

Partial correlation $\rho_{ij\cdot k} = \dfrac{\rho_{ij} - \rho_{ik}\rho_{jk}}{\sqrt{(1-\rho_{ik}^2)(1-\rho_{jk}^2)}}$

Proximity; connected; acyclic

T_1 edges become T_2 nodes

■ Known / derived ■ Uncertain / completion ■ Judgement

Vine to Matrix

Assign partial correlations

Maximise entropy – Judgement 2

Avoid unsupported dependence assumptions.

Gaussian scenario

Maximise $\det R = \prod_{e \in E(V)} (1 - \rho_e^2)$

$$\rho_{25} = \rho_{12}\rho_{15} + \rho_{25\cdot 1}\sqrt{(1-\rho_{12}^2)(1-\rho_{15}^2)}$$

$$\rho_{25\cdot 1} = 0.00 \implies \rho_{25} = \rho_{12}\rho_{15} \approx 0.22$$

Completed Pearson matrix

	X_1	X_2	X_3	X_4	X_5
X_1	1.00	−0.59	−0.46	0.31	−0.37
X_2	−0.59	1.00	0.54	−0.79	0.22
X_3	−0.46	0.54	1.00	−0.06	0.17
X_4	0.31	−0.79	−0.06	1.00	−0.12
X_5	−0.37	0.22	0.17	−0.12	1.00

Expand (Judgement 1) Assign (Judgement 2) Reconstruct

Incomplete matrix → Incomplete vine → Completed vine → Completed matrix

Vines make dependence judgements explicit and produce valid matrices by construction.

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